ProxiVideoFriends: Revisiting Proxemics through Temporal and Social Reasoning

Isabel *Jiménez-Velasco, Rafael Muñoz-Salinas, Vicky Kalogeiton and Manuel J. Marín-Jiménez*

e-mail: isajimenez@uco.es

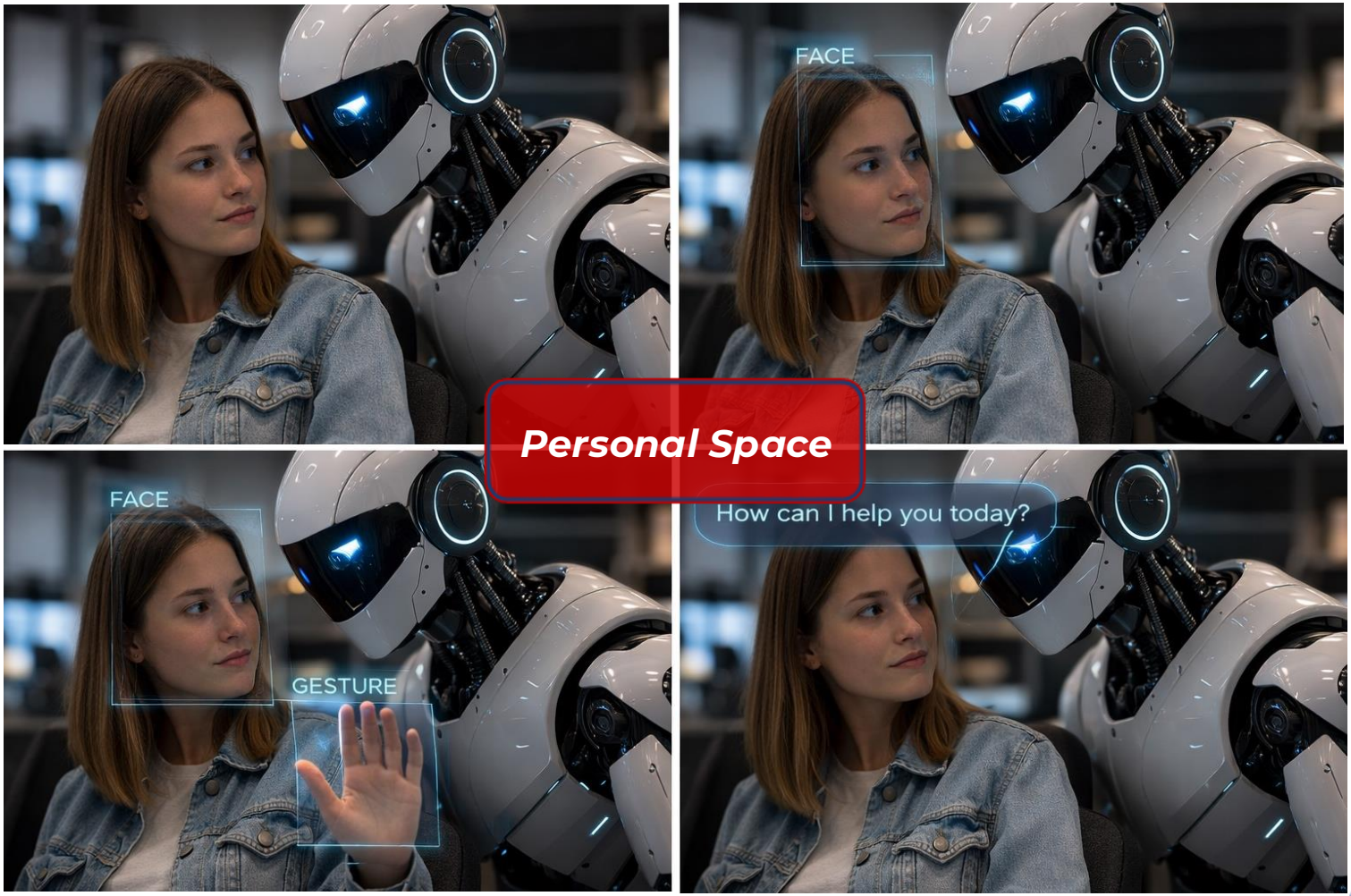
UNIVERSIDAD DE CÓRDOBA

ÉCOLE POLYTECHNIQUE UNIVERSITÉ PARIS-SACLAY

LIX

VISTA

AVA Artificial Vision Applications



Personal Space

It does not understand

how close is TOO close

Social AI must understand:

- What people **do**
- How **close** they are
- What that **spatial behavior** means

PROXEMICS

[Edward, 1963] *E. Hall. A system for the notation of proxemic behavior. American Anthropologist, 1963.*

Proxemics – Distance as social meaning

- How humans use **distance** and **body orientation.**
  - Intimicity
  - Formality
  - Dominance

- Form of **Non-verbal communication.**

Distance is not just geometry.

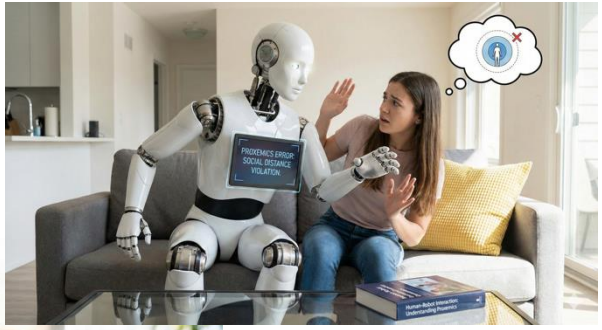
It carries social meaning.

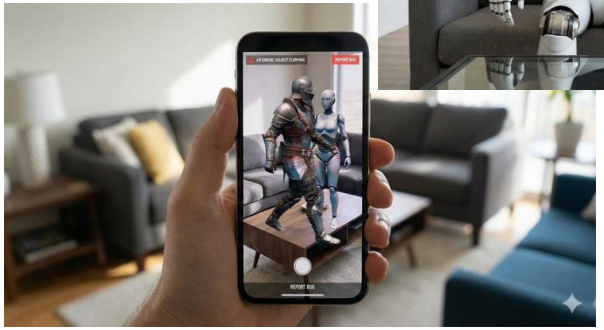
Proxemics – Why it matters

Intelligent systems are increasingly part of **daily life**.

Understanding proxemics becomes **CRITICAL.**

- Invading personal space
- Inappropriate virtual behavior
- Unrealistic simulations





Comfort. Trust.
Social realism

Are we modeling proxemics?

- Mostly – at the **image** level
- Contact **classification** from **static frames**

DYNAMIC

Contact or proximity?

Temporal disambiguation

Missing benchmark

- No dedicated **datasets** for proxemics in **video**

Temporal models cannot be systematically studied

Proxemics is social

Couple

Colleagues

Strangers

Physical distance depends on relationships

Do Visual Language Models solve this?

- Really powerful – describe scenes, reason, answer questions

Do they truly undertand fine-grained human physical contact?

Describe the image.

*A **smiling couple talks** while a TV plays in the background.*

What type of **physical interaction** do you see?

*They're **standing close** together, but there's **no physical contact**.*

Proxemics remains a blind spot

The GAP – *What is missing?*

Video benchmark

Temporal modeling

THE GAP

Spatial-Social Leaning

VLM evaluation

Physical interaction?

Standing close, *but* ***no physical contact***.

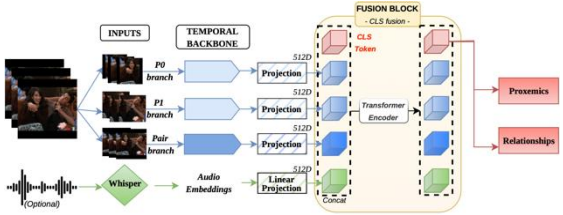
Adressing the Gap

Data Level

- *New: ProxiVideoFriends*
- First benchmark for proxemics in video

Modeling Level

- *Temporal multitask model*
- Joint learning of proxemics + relationship

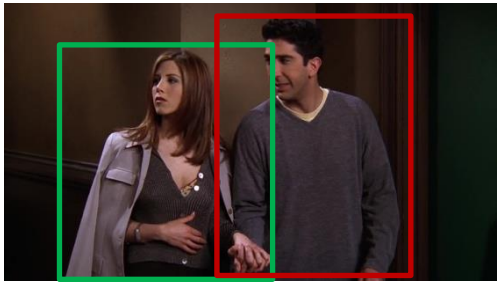


New dataset: ProxiVideoFriends

Diverse social interactions in realistic scenes

ProxiVideoFriends – Dataset overview

Per-Frame Annotation



ID: Rachel
ID: Ross

42k Frames - 103k Pairs
First video benchmark

Pair-Level Labels

Proxemics

[Yang et al., 2012]

Hand-Hand

Hand-Shoulder & Shoulder-Shoulder

Hand-Torso

Hand-Elbow & Elbow-Shoulder

Relationships

[Li et al., 2020]

Friends

Family

Couple

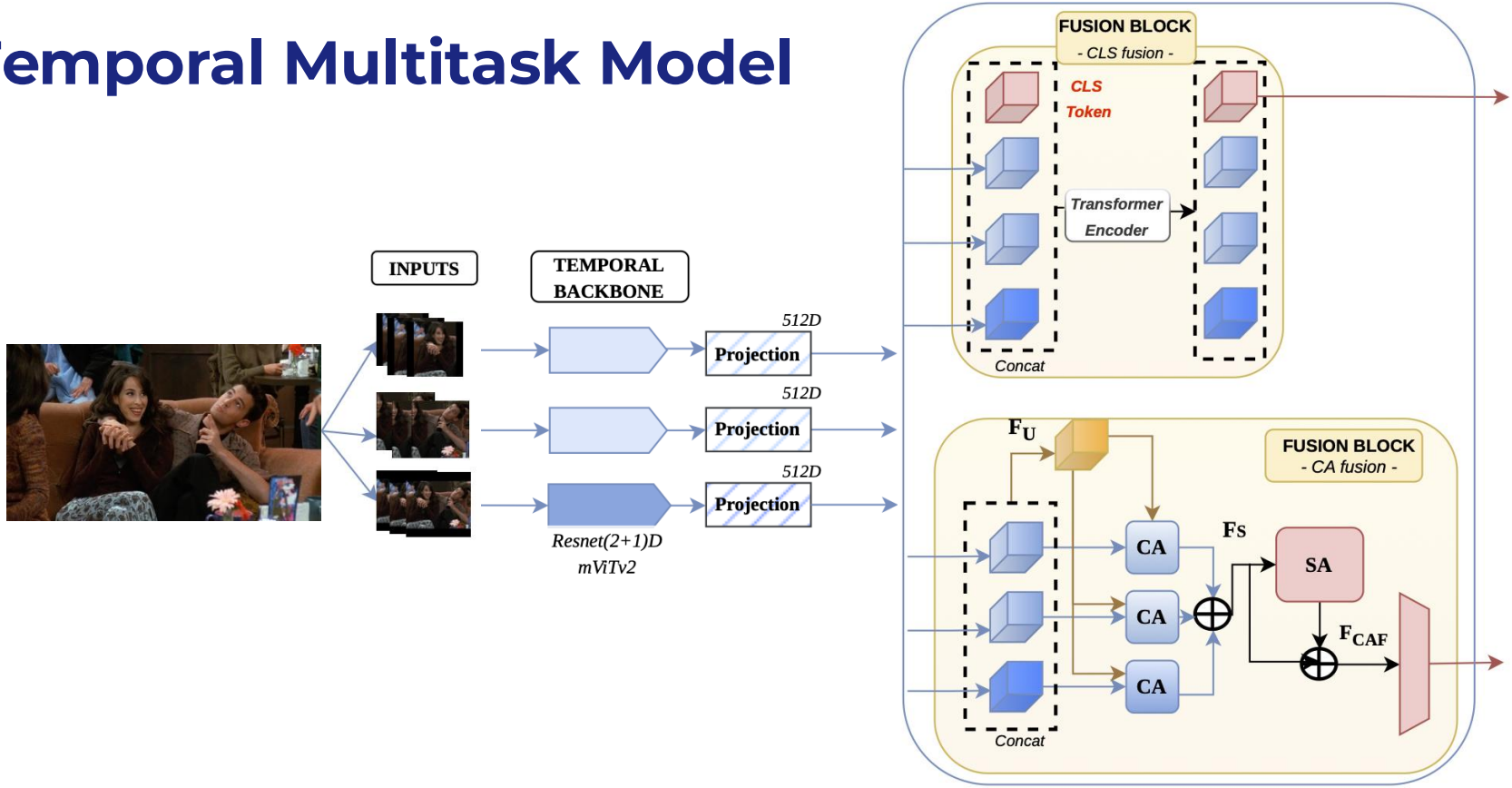
Professional

Commercial

No Relation

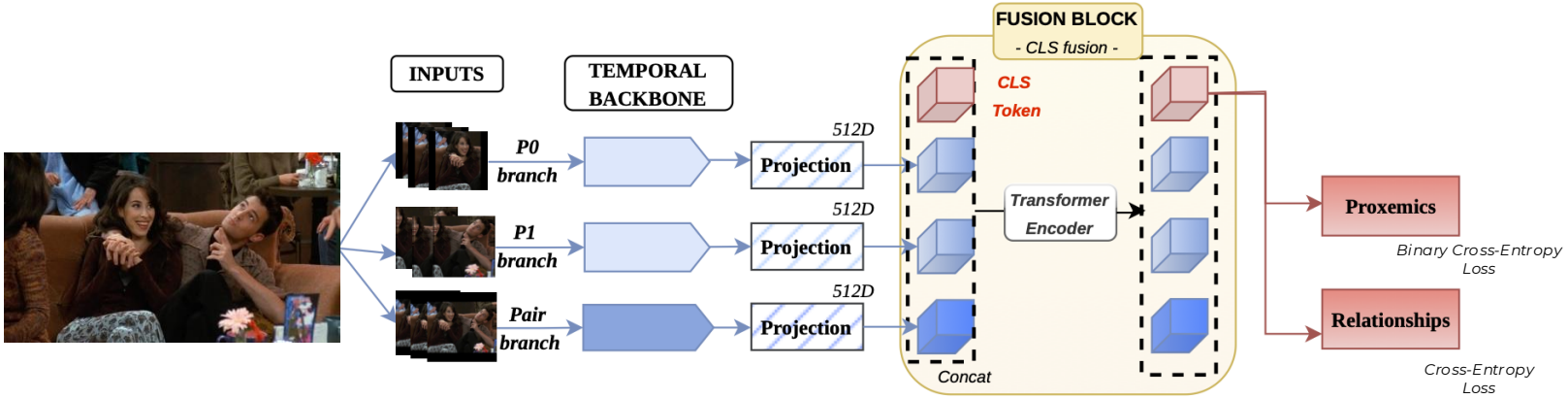
[Yang et al., 2012] Y. Yang, S. Baker, et al. *Recognizing Proxemics in Personal Photos*. CVPR, 2012.
[Li et al., 2020] J. Li, Y. Wong, et al. Visual social relationship. Recognition. IJCV, 2020.

Temporal Multitask Model



[Jimenez et. al., 2024] *I. Jiménez, J. Zafra et al. Proxemics-net++: classification of human interactions in still images. Pattern Anal. Appl., 2024.*

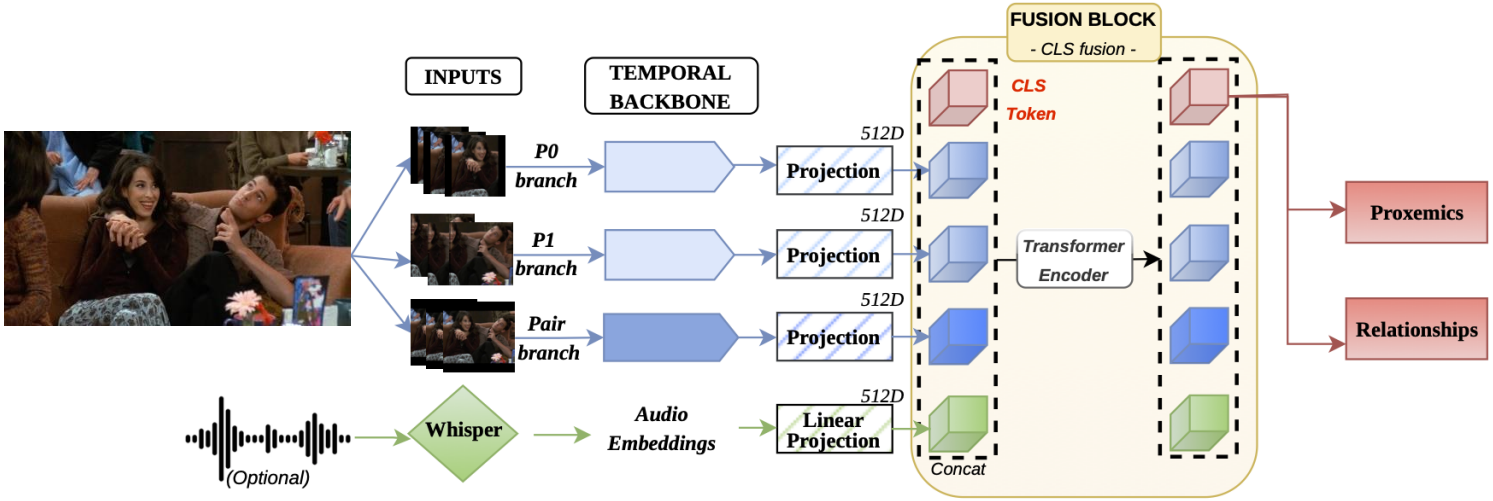
Temporal Multitask Model



Hand-Hand & Hand-Shoulder

Couple

Multitask + Multimodal Learning with Audio

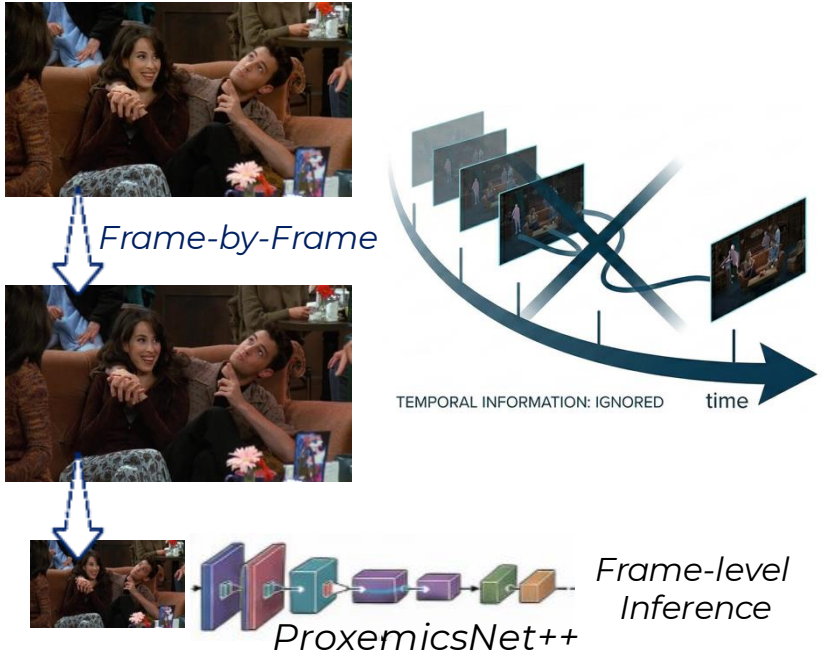


Temporal + Social + Multimodal → Proxemics in videos

Baselines - SOTA

Image-based model:

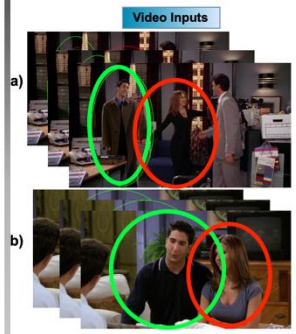
ProxemicsNet++



[Jimenez et. al.]

Video VLM:

Qwen3-VL-30B



Prompt

Given that physical interaction is defined as any form of direct bodily contact between two people, it encompasses any instance where any part of their bodies comes into physical contact.

Question: Are there any physical interactions between the two persons marked by **green and red circles**?

Possible answers choices:
A) Hand-to-Hand
B) Hand-to-Shoulder
C) Shoulder-to-Shoulder
D) Hand-to-Torso
E) Hand-to-Elbow
F) Elbow-to-Shoulder
G) None of them

Output: Answer with only the letters in the format:
"Answer: (X, Y, ...)" where X, Y are the correct digit choices.

Zero-shot
- General-purpose VLM
- No training

Out-of-the-box reasoning

Fine-tuned
- Trained on our dataset
- Supervised adaptation

Learned proxemic reasoning

Experimental Setup

Dataset Split

- **2 fold** cross-validation
- **Episode**-level Split
- **13 train** / **12 test**
- **No** visual **overlap**

Metrics

- Proxemics
  - **mAP**
- Relationship:
  - **Acc & Macro-F1**

Results: State-of-the-art

	Proxemics SOTA	mAP
Frame-level	ProxemicsNet++ (Jiménez-Velasco et al., 2024)	17.9
	VLM: Qwen3-VL (QwenTeam, 2025)	28.1
	VLM: Qwen3-VL Ft (QwenTeam, 2025)	30.6
Temporal	mViTv2 (Li et al., 2022)	30.2
	ResNet(2+1)D (Tran et al., 2018)	32.5
	Ours (Proxemics + Relationships)	**40.1**

+ 9.5%

Table: Comparison of SOTA methods for proxemics classification on ProxiVideoFriends

Take-away: Video matters - but **joint spatial + social** is the real **boost**.

Results: Is Multitask Learning Helping?

Tasks		Fusion Block		Results		
				Proxemics	Relationships	
Proxemics	Relationships	CA	CLS	mAP	Acc	F1 macro
✓	-	-	✓	32.5	-	-
-	✓	-	✓	-	39.5	17.2
✓	✓	✓	-	40.3	36.4	18.6
✓	✓	-	✓	**40.1**	**45.9**	**20.1**

Table: Performance comparison of single-task and multitask models

Take-away: *Social* ***Relationships shape Proxemics*** *behaviour.*

Results: Is Audio Helping?

	Audio	Results		
		Proxemics	Relationships	
		mAP	Acc	F1 macro
	-	**40.1**	45.9	20.1
(Ours)	✓	37.6	**46.2**	**25.0**

Table: Comparison of the impact of adding audio features to the multitask model.

Take-away: *Proxemics* *is primarily* ***visual*** *while* ***relationships*** *benefit from* ***speech*** *cues.*

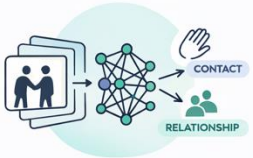
Where do we fail?

Ambiguity

Occlusion & Complexity

Takeaways

First
Video Poxemics benchmark



SOTA with
Temporal Multi-task Learning

Proxemics
Remains unsolved

Temporal **modeling is essential.**

Social **relationships improve proxemics** understanding.

Audio helps **social** reasoning – **not proxemics.**

Toward systems that understand not only what people do, but how they relate in space.

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